MILITARY SPECIFICATION

CONVERTER, VOLTAGE TO FREQUENCY
CV-3746/DKT-59

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, dev opment and test requirements for the CV-3746/DKT-59, Voltage to Frequency Converter critical item referred to herein as the converter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specifi the following specifications, standards, and handbooks of the issue listed that issue of the Department of Defense Index of Specifications and Standa (DoDISS) specified in the solicitation, form a part of this specification the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any perti-nent data which may be of use in improving this document should be addres to: Commanding Officer, Naval Air Engineering Center, Systems Engineerin and Standardization Department (SESD) Code 93, Lakehurst, New Jersey 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-T-85650	Transmitting Set, Telemetric Data AN/DKT-59
MIL-A-85652	Amplifier, Mixer AM-7157/DKT-59

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics

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MIL-STD-781	Reliability Design Qualification and Product Acceptance Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-883	Test Methods and Procedures for Microelectror
MIL-STD-965	Parts Control Program
MIL-STD-2078	Requirements for Preparation of Support Equipment Depot Level Rework Standards
MIL-STD-45662	Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

1356AS310	Converter, Voltage to Frequency

PUBLICATIONS

INTER RANGE INSTRUMENTATION GROUP (IRIG) STANDARD 106-80	Telemetry Standards

(Application for copies should be addressed to the Department of Defense Technical Information Center, Cameron Station, Alexandria, VA 22314.)

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Converter description. The converter converts high level input signal voltage into a frequency modulated (FM) subcarrier and when utilized with one or more converter modules, each at a different center frequency, provides an FM/FM type signal to the AM-7157/DKT-59-amplifier mixer, MIL-A-85652(AS).

3.1.1 Interface definition. The converter's physical and electrical interfaces with the module assembly unit shall be in accordance with Drawing 1356AS310.

3.2 Characteristics.

3.2.1 Performance. The converter shall meet the performance requirements specified herein under the conditions of 3.2.5 (see 4.6.2.3 and 4.6.2.9).

3.2.1.1 External power. The converter shall operate using external power with characteristics as specified in MIL-STD-704 and modified herein as follows:

a. 20.0 to 36.0 volts direct current (Vdc)

b. Transient surge from +40 to -40 Vdc

c. Maximum current shall be not greater than 0.010 amperes

3.2.1.2 Input characteristics. The converter shall have the following input characteristics:

a.	Input impedance:	Shall be not less than 500 Kilohms (K ohms)(see 4.6.2.9.1b).
b.	Input signal range:	0.0 to 5.0 Vdc (with mode adjust input grounded) or -2.5 to +2.5 Vdc (with mode adjust input ungrounded) (see 4.2.6.9.1b).

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3.2.1.3 Modulation requirements. The converter shall meet the modulati requirements as follows:

a. Modulation sensitivity:	0.0 to 5.0 Vdc input or -2.5 to +2.5 Vdc input (with mode pin ungrounded) for -7.5 to +7.5 percent deviation of the center frequency (f_o) (see 4.6.2.9.1c).
b. Modulation sense:	Increasing input voltage in the positive direction shall produce an increase in output frequency (see 4.6.2.9.1d).
c. Source impedance sensitivity:	Changing the source impedance from zero to infinity shall vary the frequency less than 1.5 percent of desi bandwidth for a source voltage of zero (see 4.6.2.9.1e).
d. Frequency range:	Shall be in the range of the applica proportional channel band defined in IRIG 106-80. Specific IRIG channels will be designated by the procuring activity (see 4.6.2.7 and 6.2.1).
e. Intelligence frequency response:	Shall be not greater than +0.2 decibe (dB) of direct current (dc) response for a modulation index of 1.0 or greater (see 4.6.2.9.1f).
f. Output impedance:	47K ±4.7K ohms (see 4.6.2.9.1g).
g. Output impedance sensitivity:	A 10 to 1 change in the load impedanc shall cause less than 0.5 percent of design bandwidth change in the output frequency (see 4.6.2.9.1h).

3.2.1.4 Center frequency deviation. The center frequency deviation of the converter shall meet the limits as follows:

a. The deviation shall be ±7.5 percent of center frequency to an accuracy of ±2 percent of design bandwidth for input signals of 0.0 and 5.0 Vdc, or -2.5 and +2.5 Vdc, at ambient temperatures (see 4.6.2.2).

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b. The deviation change shall be not greater than 0.5 percent of design bandwidth for an external power as specified in 3.2.1.1a (see 4.6.2.2).

3.2.1.5 Output voltage. The output voltage of the converter shall be 325 ±65 millivolts (mV) for all combinations of environment and power levels at center frequency measured across a 10K ohm load (see 4.6.2.3).

3.2.1.6 Linearity. The characteristic curve of output frequency versus input signal shall be linear within 0.25 percent of design bandwidth for ±7.5 percent f_0 deviation. There shall be no limiting, flat spots, or noise spikes. Linearity shall be determined as the deviation from the best straight line by the least squares method for all input signal and frequency variations (see 4.6.2.3).

3.2.1.7 Repeatability. The repeatability error of the converter shall be not greater than 1 percent of the design bandwidth for all input signals and assigned channel bands for a period of eight hours at ambient temperature (see 4.6.2.5).

3.2.1.8 Amplitude modulation. The amplitude modulation on the output of the converter shall be not greater than five percent for all input signal and frequency deviations (see 4.6.2.6).

3.2.1.9 Intelligence frequency component. The intelligence frequency component on the output signal of the converter shall be not greater than 40 dB below the output for all input power and nominal frequency variations as specified in IRIG 106-80 for that channel (see 4.6.2.7).

3.2.1.10 Harmonic distortion. The harmonic distortion on the output of the converter shall be not greater than 0.75 percent for all input signal and frequency deviations (see 4.6.2.8).

3.2.1.11 Transient surge voltage protection. The converter shall recover automatically to performance as specified herein following a transient surge dc voltage step function as specified in 3.2.1.1 (see 4.6.2.9.2c).

3.2.1.12 Temperature stability. At any temperature from -40 degrees centigrade (°C) to +71°C, the output frequency shall not change more than ±2 percent of design bandwidth when the input is constant anywhere within the 0 to 5 Vdc or -2.5 to +2.5 Vdc range from a 25°C reference (see 4.6.2.10).

3.2.2 Physical characteristics. The converter shall meet the physical requirements of Drawing 1356AS310.

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3.2.3 Reliability. The reliability of the converter shall be not les than 2000 hours mean-time-between-failures (MTBF).

3.2.4 Maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability paramet shall be as specified in MIL-STD-2078.

3.2.5 Environmental conditions.

3.2.5.1 Operational. The converter shall meet all requirements of 3.2 and 3.2.3 prior to, during, and after exposure to the following environme conditions, separately or any combination thereof. The converter shall n exhibit any degradation in excess of limits specified herein as the resu of exposure to any of the environments in a through f of Table I (see 4.6

3.2.5.2 Non-operational. The converter shall meet all requirements of 3.2.1 and 3.2.3 after exposure to the environmental conditions c, g, and of Table I, separately or any combination thereof. The converter shall n exhibit any temporary or permanent degradation in excess of limits specif herein as the result of exposure to any of the environments specified abo (see 4.6.3).

TABLE I. Environmental conditions.

PARAMETER	OPERATIONAL	NON-OPERATIONA
a. Pressure	70,000 ft (equivalent pressure)	-----
b. High Temperature	+71°C	-----
c. Low Temperature	-40°C	-40°C
d. Acceleration	40 g, 3 axes	-----
e. Random Vibration	20-2000 Hz, .2g^2/Hz	-----
f. Shock	65g, 5ms, 1/2 sine	-----
g. Thermal Shock	-----	-55°C to +85°C
h. Moisture Resistance	-----	95% relative humidity

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3.2.6 Storage life. The storage life of the converter shall be such that it can be stored for three years in the shipping or storage container and then meet the requirements specified herein.

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3.1 Visual examinations. Test samples shall be subjected to visual examination to verify conformance to the physical and interchangeability requirements of this specification and the contract (see 4.6.1.).

3.4 Design and construction. Design and construction shall be in accordance with Drawing 1356AS310 and requirements specified herein (see 4.6.1).

3.4.1 Materials, processes, and parts. Fulfillment of major design objectives for the intended application specified herein shall be the prime consideration in the selection of materials, processes, and parts. In so doing, the following shall govern (see 3.4.1 and 4.6.1.2):

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein (see 4.6.1.2).

b. Other materials, processes, and parts requirements shall conform to MIL-E-5400 (see 4.6.1.2).

c. Parts shall be selected using the following order of preference (see 4.6.1.2).

(1) Parts listed on a Qualified Parts List (QPL) with two or more sources.

(2) Parts listed on a QPL with only one source.

(3) Commercial parts with two or more sources.

(4) Commercial parts with only one source.

3.4.1.1 Standard parts. Parts specified in 3.4.1c (1) and (2) are standard parts and do not require approval from the procuring activity. Parts selected for use from 3.4.1c (3) and (4) are considered nonstandard and require approval. Nonstandard parts approval requests shall be submitted on DD Form 2052 in accordance with the contract (see 6.2.2).

3.4.1.2 Commercial parts. Commercial parts specified in 3.4.1c (3) and (4) shall be readily available, nationally catalogued, off-the-shelf items.

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3.4.1.3 Nonstandard parts. Use of nonstandard parts in first article production units prior to Government approval is at the sole risk of the contractor and subject to replacement if the Government does not approve proposed nonstandard part.

3.4.2 Electromagnetic interference.

The converter shall meet the electromagnetic interference (EMI) requirements of MIL-STD-461, tests CE03, CE07, CE01, CS01, CS02, CS06, RE02, RS02 and RS03.

3.4.3 Nameplate and marking. The nameplate shall be as specified in Drawing 1356AS310. Interchangeable parts and assemblies shall be legibly marked. All markings shall be as specified in MIL-STD-130 (see 4.6.1.3).

3.4.4 Serial numbers. Serial number assignments shall be as specified MIL-N-18307 (see 4.6.1.3 and 6.2.2).

3.4.5 Interchangeability. All assembly and subassembly level items of hardware that are subject to replacement and have the same part number sha be electrically, physically, and functionally interchangeable. Electrical component parts shall conform to the requirements specified in MIL-STD-965 The converter shall meet the interchangeability requirements of MIL-STD-45 Requirement 7, and the electrical and mechanical interface requirements of the Telemetry Data AN/DKT-59 Transmitting Set (MIL-T-85650)(see 4.6.2.11).

3.4.6 Safety. The converter shall meet the safety requirements of MIL-STD-454, Requirement 1.

3.4.7 Human engineering. All controls and adjustments shall be set dur production and initial testing and shall be inaccessible in the field.

3.5 Soldering. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5 (see 4.6.1.4).

3.6 Workmanship. Fabrication and workmanship practices shall be in acc dance with MIL-STD-454, Requirement 9 (see 4.6.1.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the co tract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to monitor any of the inspections or to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections and tests specified herein. The contractor's Quality Assurance Program shall be planned and conducted in a manner to achieve levels of quality and reliability commensurate with the intended application specified herein.

4.1.1.1 Inspection. The contractor's inspection system shall conform to MIL-STD-883 for both the commercial off-the-shelf microelectronics and new contractor-designed (peculiar) microelectronics.

4.1.1.2 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Test plans and procedures. Detailed test plans and procedures in accordance with MIL-T-18303 shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract (see 6.2.1). When approved test procedures are available from previous contracts, such procedures will be provided and may be used when approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare test data showing quantitative results for all first article and quality conformance inspections (see 6.2.2) for each converter offered for acceptance. This report shall also identify all rejections from tests.

4.1.3.1 Electromagnetic interference test data. The contractor shall prepare a electromagnetic interference test report in accordance with MIL-STD-831 (see 6.2.2).

4.1.4 Failure reports. All failures shall be documented in a failure report, identifying the cause of failure and the proposed corrective action for each failure (see 6.2.2).

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4.1.5 Reliability Reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability tes data, including failure reports, in accordance with MIL-STD-781 (see 6.2.2)

4.2 Classification of inspections. The inspection requirements specifie herein are classified as follows:

a. First article inspection (see 4.4)

(1) Visual examination (see 4.6.1)

(2) Performance test (see 4.6.2)

(3) Environmental test (see 4.6.3)

(4) Reliability test (see 4.6.4)

b. Quality conformance inspection (see 4.5)

(1) Individual test (see 4.5.2)

(2) Periodic production test (see 4.5.3)

4.3 Inspection conditions. Unless otherwise specified herein, all inspec tions shall be performed in accordance with the test conditions specified in 4.3.3.

4.3.1 Maintenance and calibration. All test equipment shall be calibrate and maintained in accordance with MIL-STD-45662. Certification to this effe shall accompany all reports of first article and quality conformance tests (see 6.2.2).

4.3.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the toleranc for the variable to be measured. In those cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within five percent of that requirement.

4.3.3 Test environment. Unless otherwise specified in the applicable tes procedure, all tests and inspections shall be conducted under the following environmental conditions:

a. Temperature (ambient): +23 ±10°C

b. Altitude: elevation of test facility

c. Relative humidity: 50 ±30 percent

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4.3.4 Tolerances on environmental test conditions. The allowable tolerances on environmental test conditions shall be in accordance with MIL-T-5422.

4.4 First article inspections. When specified in the contract (see 6.2.1), first article inspections shall be conducted by the contractor on a converter representative of the production converters to be supplied under the contract. The first article sample (see 3.3) shall be subjected to the examinations and tests of Table II and shall be performed in the sequence specified therein.

4.4.1 Acceptance criteria. Failure of any converter to pass any first article test shall result in discontinuance of testing and rejection of the first article sample. The procuring activity reserves the right to require an additional sample and test if any converter fails to meet any requirement of this specification.

4.4.1.1 Post-tests. A converter which has previously failed first article inspection shall be repaired and retested in accordance with 4.6.2 prior to its resubmission for first article acceptance.

4.4.2 First article failure analysis and corrective action. The procuring activity shall be notified of a first article test failure and its probable cause (see 4.1.4 and 6.2.2).

4.5 Quality conformance inspections. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspection and testing shall be subject to monitoring by the Government inspector. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual tests (see 4.5.2)

b. Periodic production tests (see 4.5.3)

4.5.1 Quality conformance acceptance criteria. Any converter that fails to pass any individual test shall be rejected.

4.5.2 Individual tests. The individual tests shall be conducted on each converter offered for acceptance and shall consist of the tests in Table II.

4.5.3 Periodic production tests. After the first 250 converters are produced, a periodic sample consisting of one unit shall be selected at random from the first 10 converters of every 250 produced thereafter. This sample shall be selected from converters which have previously passed the individual

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TABLE II. Individual tests.

Examination or Test*	Requirement Paragraph	Method Paragraph		
		First Article	Quality Conformanc	
			Individual	Period Product
Visual examination	3.3.1	4.6.1	4.6.1	4.6.1
Performance tests				
Electromagnetic interference	3.4.2	4.6.2.1		
Center frequency deviation limit	3.2.1.4	4.6.2.2	4.6.2.2	4.6.2.2
Output voltage	3.2.1.5	4.6.2.3	4.6.2.3	4.6.2.3
Linearity	3.2.1.6	4.6.2.4		4.6.2.4
Repeatability	3.2.1.7	4.6.2.5		4.6.2.5
Amplitude modulation	3.2.1.8	4.6.2.6		4.6.2.6
Intelligence frequency component	3.2.1.9	4.6.2.7		4.6.2.7
Harmonic distortion	3.2.1.10	4.6.2.8	4.6.2.8	4.6.2.8
External power voltage variations	3.2.1.1	4.6.2.9		4.6.2.9
Input characteristics	3.2.1.2	4.6.2.9.1b		4.6.2.9.1
Modulation requirements	3.2.1.3	4.6.2.9.1c		4.6.2.9.1
Output voltage	3.2.1.5	4.6.2.9.2b		4.6.2.9.2
Transient surge voltage protection	3.2.1.11	4.6.2.9.2c		4.6.2.9.2
Temperature stability	3.2.1.12	4.6.2.10	4.6.2.10	4.6.2.10
Interchangeability	3.4.5	4.6.2.11	4.6.2.11	4.6.2.11
Environmental test	3.2.5	4.6.3		4.6.3
Reliability test	3.2.3	4.6.4		4.6.4

*All examinations and tests shall be performed in top to bottom sequence.

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tests of 4.5.2. The sample shall be subjected to all the examinations and tests specified in Table II.

4.5.3.1 Periodic production sample rejection. If the periodic production sample fails to meet the requirement of any examination or test specified herein, the sample shall be rejected. Before any further production lots or converters may be submitted for acceptance, a detailed report shall be prepared (see 6.2.2) describing the rejection and action taken to prevent recurrence of the defect which caused the failure. Further production lots will not be considered for acceptance until the periodic production sample has been approved.

4.6 Examination and test methods. Details of methods and procedures shall be in accordance with the approved test plan (see 4.1.2).

4.6.1 Visual examination.

4.6.1.1 Physical characteristics. The converter shall be inspected visually for conformance to 3.2.2.

4.6.1.2 Materials, processes, and parts. The converter shall be examined visually for conformance to 3.4.1.

4.6.1.3 Nameplates and product marking. The converter shall be inspected for conformance to 3.4.3.

4.6.1.4 Soldering. The converter shall be examined for conformance to 3.5.

4.6.1.5 Workmanship. The converter shall be examined for conformance to 3.6.

4.6.1.6 Human engineering. The transmitting set shall be examined to verify conformance to the requirements of 3.4.7.

4.6.1.7 Interchangeability. The converter shall be examined to verify physical interchangeability conformance to 3.4.5.

4.6.2. Performance tests. Performance tests shall be performed as specified herein.

4.6.2.1 Electromagnetic interference. The converter sections shall be tested for conformance to 3.4.2 and in accordance with MIL-STD-462.

4.6.2.2 Center frequency deviation limit. The converter center frequency deviation limit shall be measured. The external power, input signal, output frequency, and load shall be as specified in 3.2.1.1, 3.2.1.2, 3.2.1.4, and 3.2.1.5.

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4.6.2.3 Output voltage. The converter output voltage shall be measured. The external power, input signal, output voltage, and load shall be as specified in 3.2.1.1, 3.2.1.2, 3.2.1.4, and 3.2.1.5.

4.6.2.4 Linearity. The converter linearity shall be measured at five points of the design bandwidth from -7.5 percent f_o to +7.5 percent f_o. The input power, input signal, and linearity at the converter output shall meet the requirements of 3.2.1.1, 3.2.1.2, and 3.2.1.6.

4.6.2.5 Repeatability. Converter repeatability measurements shall be taken at 30 minute intervals and at one and four volts input signal for eight hours at constant ambient temperature. The repeatability of the converter shall meet the requirements of 3.2.1.7.

4.6.2.6 Amplitude modulation. The converter amplitude modulation shall be measured while applying a 20 Hz, zero to five volt peak-to-peak input signal. The amplitude modulation characteristics shall meet the requirement of 3.2.1.8.

4.6.2.7 Intelligence frequency component. The intelligence frequency component shall be measured with an input signal that is a zero to five volt peak-to-peak sine wave whose frequency is equal to the "nominal frequency response" specified in IRIG 106-80 for that channel. The intelligence frequency component shall meet the requirements of 3.2.1.9.

4.6.2.8 Harmonic distortion. The converter's harmonic distortion shall be measured at center frequency while applying a 2.5 Vdc input signal (mode adjust grounded) and external power as specified in 3.2.1.1. The harmonic distortion shall meet the requirements of 3.2.1.10.

4.6.2.9 External power voltage variation. External power voltage variation shall be as specified in 4.6.2.9.1 and 4.6.2.9.2.

4.6.2.9.1 Steady-state voltage test. External voltage shall be applied along with appropriate signal inputs to the converter. The external voltage shall be adjusted to +20.0 +0.25,-0.0 Vdc and the following verified:

a. Power demand as specified in 3.2.1.1

b. Input characteristics as specified in 3.2.1.2

c. Modulation sensitivity as specified in 3.2.1.3(a)

d. Modulation sense as specified in 3.2.1.3(b)

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e. Source impedance sensitivity as specified in 3.2.1.3(c)

f. Intelligence frequency response as specified in 3.2.1.3(e)

g. Output impedance as specified in 3.2.1.3(f)

h. Output impedance sensitivity as specified in 3.2.1.3(g)

With external voltage of +36.0 +0.0,-0.25 Vdc applied, the test shall be repeated.

4.6.2.9.2 Transient voltage test. The converter shall be operated with external voltage of +28.0 ±0.5 Vdc for a period of three minutes while monitoring output voltage and center frequency. The external voltage shall be reduced to zero for not less than 1 second, raised to +40 ±1.0 Vdc for not less than 10 seconds, reduced to -40 ±1.0 Vdc for 10 seconds then returned to +28.0 ±0.5 Vdc and the following verified:

a. Center frequency deviation as specified in 3.2.1.4

b. Output voltage as specified in 3.2.1.5

c. Transient surge voltage protection as specified in 3.2.1.11.

4.6.2.10 Temperature stability. The converter temperature stability shall be measured at -40°C, 25°C, and 71°C and shall meet the requirements of 3.2.1.12.

4.6.2.11 Interchangeability. The converter shall be tested to verify electrical and operational interchangeability conformance to 3.4.5.

4.6.3 Environmental tests. Environmental tests shall be performed as specified herein. When tests are conducted at high or low temperatures, test converters shall be stabilized at the specified temperature for not less than two hours prior to testing. If the converters are removed from the chamber for any portion of the tests, such tests shall be completed within 10 minutes of removal from the chamber. If the tests are not completed within the 10 minute period, the converters shall be returned to the temperature chamber for not less than two hours before continuing the tests. The test converters shall stabilize at room ambient conditions prior to performance of post tests and measurements.

4.6.3.1 Operational environmental tests. The converter performance shall be tested as specified in 4.6.3.1.1 through 4.6.3.1.6 and meet the requirements of 3.2.1.1 through 3.2.1.12 during and after exposure to the environmental conditions specified in Table I, a through f.

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4.6.3.1.1 Reduced pressure. The converter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.2 High temperature. The converter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.3 Low temperature. The converter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 Equipment Class 2.

4.6.3.1.4 Acceleration. The converter shall be tested in accordance wit 3.2.5.1 and Requirement d. of Table I.

4.6.3.1.5 Random vibration. The converter shall be tested with a spectr power density of 0.2 g^2/Hz from 20 to 2000 Hz for 2 hours in each of three mutually perpendicular axes.

4.6.3.1.6 Shock. The converter shall be tested in accordance with MIL-T-5422, Shock Test.

4.6.3.2 Nonoperational environmental tests. The converter's performance shall be tested as specified in 4.6.3.2.1 through 4.6.3.2.3 and meet the requirements of 3.2 after exposure to the environmental conditions specifie in Table I, c, g, and h.

4.6.3.2.1 Low temperature. The converter shall be tested in accordance with MIL-T-5422, Temperature and Altitude Tests for MIL-E-5400 equipment Class 2 at -40°C for 48 hours.

4.6.3.2.2 Thermal shock. The converter shall be tested in accordance with MIL-T-5422, Temperature Shock Test.

4.6.3.2.3 Moisture resistance. The converter shall be tested in accordance with MIL-T-5422, Humidity Test.

4.6.4 Reliability test. The converter shall be tested as specified in the Probability Ratio Sequential Test (PRST) Plan VIIIC, Figure C-8, Appendix C of MIL-STD-781 to demonstrate compliance with 3.2.3.

4.6.4.1 Reliability acceptance criteria. The Government inspector shall be notified any time the total test hours and test failures plotted in accordance with the PRST Acceptance and Reject criteria for Test Plan VIIIC of MIL-STD-781 reflect a "reject" situation during the conduct of the contractor's approved production test procedures. The procuring activity

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reserves the right to stop the acceptance of products at any time a reject situation exists pending a review of the contractor's efforts to improve the product, product components and parts and the product workmanship.

4.7 Production converters. Converters supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article. Each converter shall be capable of successfully passing the same tests as imposed on the first article. Evidence of noncompliance with the above shall constitute cause for rejection, and, for converters already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections.

4.8 Packaging inspection. The preservation, packing and marking of the converter shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation-packaging, and packing shall be as specified in MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for Level C shall be determined in accordance with the selection chart in Appendix D specified in MIL-STD-794 and the method of packaging shall be as specified in MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3 Spare parts. When specified (see 6.2.1), spare parts shall be preserved and packaged to the same specifications as a complete converter product except that weight of spare parts will determine selection of style of the box or container.

6. NOTES

6.1 Intended use. The CV-3746/DKT-59 Voltage To Frequency Converter is intended for use in the AN/DKT-59 Telemetric Data Transmitting Set.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify following:

a. Number, title, and date of this specification.

b. Number of converters to be procured.

c. First article inspection if different from 4.4.

d. Test plans and procedures if different from 4.1.2.

e. Facility assigned to perform inspections (see 4.1).

f. Test data and reports if different from 4.1.3.

g. Acceptance criteria if different from 4.4.1 and 4.5.1.

h. Inspection conditions if different from 4.3.

i. Test equipment accuracy if different from 4.3.2.

j. Level of packing and packaging if different from 5.1.

k. Special marking (see 5.2.1).

l. Spare parts preservation and packaging (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquis tion which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and deliver in accordance with the approved CDRL incorporated into the contract. Whe the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor i accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph number	Data requirements	Applicable DID number	Opti
4.1.2	Procedures, Test	UDI-T-21347	---
4.1.3	Reports, Test	DI-T-2072	---

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4.1.3.1	Electromagnetic Interference Test Report	DI-R-7062	----
4.1.4	Report, Failed Item Analysis	DI-R-7039	----
4.1.5	Reports, Reliability Test and Demonstration	DI-R-7034	----
4.3.1	Equipment Calibration Procedures	DI-R-7065	----
4.4.2	Failure Analysis and Corrective Action Report (First Article)	DI-R-5299	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the first article sample.

Preparing activity:
Navy - AS
(Project No. 5815-N116)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85651(AS)	Converter, Voltage to Frequency CV-3746/DKT-59

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

CUT ALONG THIS LINE.)

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